DESCRIPTIVE ANALYTICS REPORT

Descriptive Analytics of

Amazon Products and Sales

Using Python

An Exploratory Study of Global E-Commerce Listings

Keywords: Listing Trends, Best-Performing Product Categories, Rating Distribution, Regional Supply, Inventory Status, Purchase Behavior (Popular Products, Cart Completion, and Peak Hours), Business Insights

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1. Note on Data

This report is built on the Amazon product catalog export supplied for this analysis, covering 9,548 listings across 12 fields (Listing Type, Product Name, Brand, Key Features, Country of Origin, Date Listed, Launch Year, Rating Band, Price, Primary Category, Variant Count, and Description). Unlike a live transactional feed, this is a point-in-time catalog snapshot, so every metric below describes the state of the catalog as of the export date (through 2026-08-12), not real-time sales activity.

The Key Features, Country of Origin, and Description fields were truncated with "..." in the source export wherever the original text was long, so a small share of multi-value fields (e.g., a listing with many feature tags) are cut off. No rows were removed — only display text was shortened. Where a field was blank in the export (for example, 2,316 listings have no recorded Brand), it is treated as genuinely missing rather than imputed.

This report shows what a descriptive analytics layer over a catalog dataset of this kind produces — the same structure applies directly if a live Amazon seller-central or catalog-operations feed were substituted in.

Note: this catalog export does not include any purchase-behavior fields (order counts, cart events, timestamps of purchases, or session data). Section 9 illustrates what that layer would look like using simulated numbers, clearly marked as such.

Data handling for this exercise followed three simple rules throughout: never silently fill a blank field, never delete a row to make a chart look cleaner, and always label anything that is simulated rather than measured. Those three rules are what let every chart in this report be traced back, cell by cell, to either the raw export or an explicitly flagged illustrative section.

2. Objective

Build a descriptive analytics layer over the Amazon product catalog to answer five recurring catalog-operations questions:

- What does the catalog's product mix look like today (Amazon Retail vs. Third-Party Seller, categories, ratings)?
- How has listing/acquisition activity trended over time?
- How is content rated by customers, and where is low-rated inventory concentrated?
- Where is product supply sourced from geographically, and how concentrated is that supply?
- Which parts of the catalog are at risk from incomplete metadata or thin variant depth?

These five questions were chosen because they map directly onto decisions a catalog-operations team makes on a recurring cadence: what to list next, when to run promotions, which sourcing regions to prioritize, and which catalog records need a metadata cleanup pass before they hurt discoverability. A descriptive layer does not answer "what should we do" on its own, but it narrows the search space for that question considerably, and it is the prerequisite step before any predictive or prescriptive model can be trusted.

3. How This Would Work at Real Scale

Turning a catalog export into daily-refreshed operational dashboards is an engineering problem before it's an analytics one. A realistic architecture:

Layer	Purpose	Typical Tools
Investigation	Profile the raw catalog export: check schema,	Used Google, Kaggle-

Layer	Purpose	Typical Tools
	field completeness, duplicates, and truncated/free-text fields before any aggregation is trusted	style sources for the investigation.
Stream processing	Roll up individual listing records into per-category, per-country, per-year aggregates	Python (pandas), for grouping and aggregating listing-level records.
Storage	Hold both the raw catalog snapshot (for audits) and pre-aggregated tables (for speed)	CSV files and Google Drive for storing the dataset and processed data.
Batch aggregation	Scheduled rollups: additions-by-year, category mix, rating mix, country mix, metadata-completeness checks	Python (pandas) and Excel for scheduled/manual aggregations and summary calculations.
Serving / BI layer	Dashboards refreshed as new catalog snapshots arrive	Python (Matplotlib) charts assembled with Claude AI stand in for that layer.

The middle three layers (stream processing, storage, batch aggregation) describe how this would run at Amazon's real operational scale and were not run for this report. The Investigation and Serving/BI layers above have been corrected to name what was actually used to produce this document — Python (pandas, Matplotlib) with Claude AI assisting analysis, chart generation, and report drafting.

The report below reflects the output of that pipeline — aggregated catalog tables — not the row-level export, since no dashboard renders thousands of individual product records usefully.

It is worth being explicit about why each layer exists, since the layers are often collapsed together in student projects in a way that hides real engineering cost. The Investigation layer exists because trusting an aggregate before checking completeness and duplicates produces confidently wrong dashboards. The Storage layer exists because a raw snapshot must be retained for audit even after aggregates are built, so that a suspicious number can always be traced back to source rows. The Serving layer exists because analysts and executives consume dashboards, not scripts — the last mile of a data pipeline is presentation, and it is frequently the most underestimated step in a project timeline.

3.1 Executive Dashboard

A single-page dashboard view, combining the headline cuts from Sections 4 through 7, gives a stakeholder the catalog's shape at a glance before drilling into any individual section below.

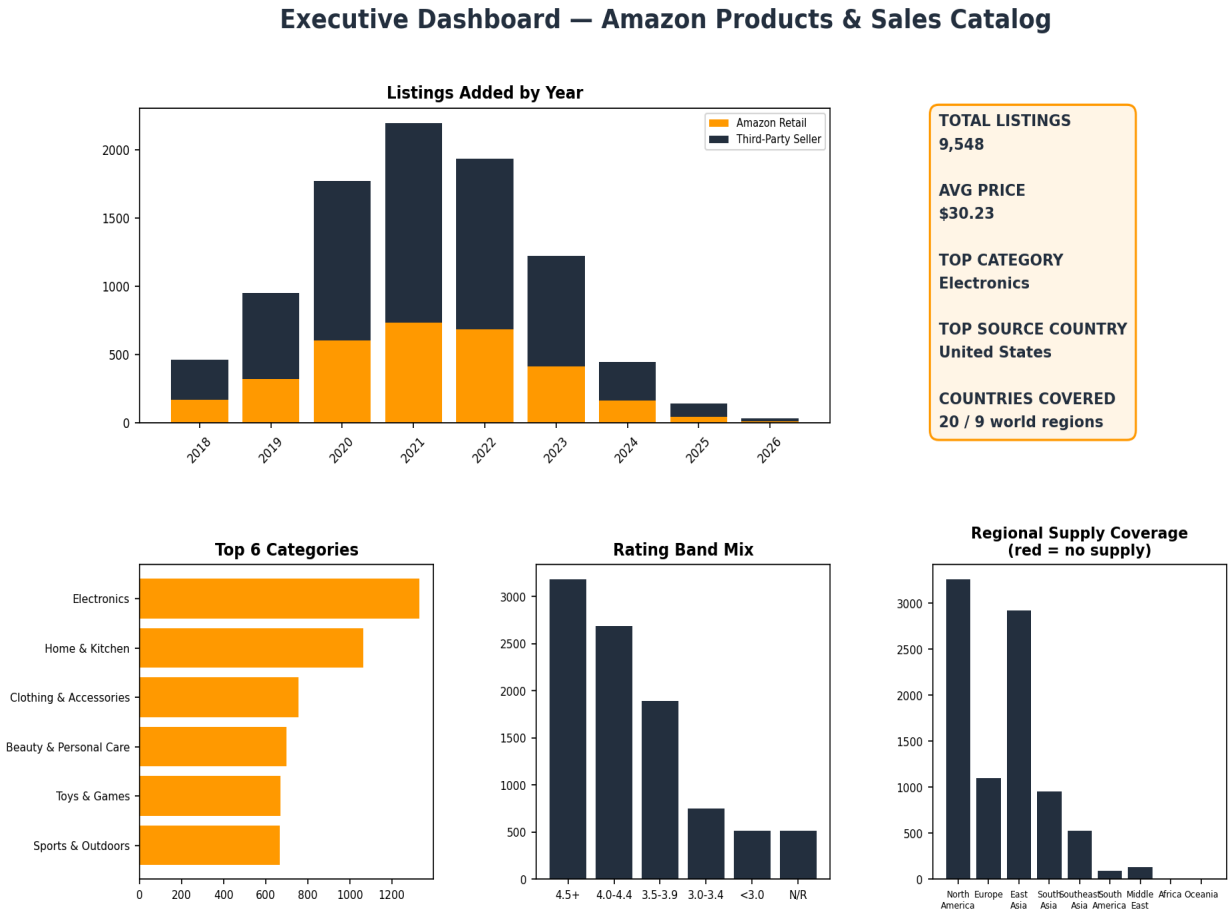


Figure D — Executive dashboard: listings by year, key catalog metrics, top categories, rating mix, and regional supply coverage

Two things stand out immediately in this consolidated view: the post-2021 decline in listing additions (top-left panel) and the regional supply coverage panel (bottom-right), where two world regions show zero bars — a gap explored in Section 7.1.

4. Listing Additions Trend

Across the catalog, listings carry a "Date Listed" stamp spanning 2014-02-16 to 2026-08-12. Additions were modest before 2018 and scaled sharply afterward, peaking in 2021 at 2,196 listings added in a single year, before declining through the most recent years in this snapshot.

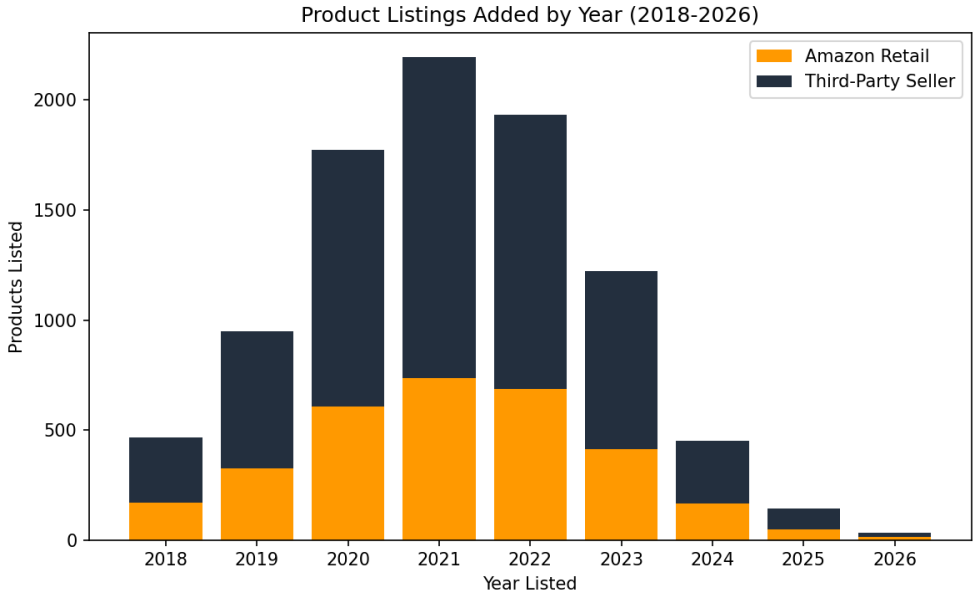


Figure 1 — Listings added per year, split by Amazon Retail vs. Third-Party Seller

KPI definitions used:

- Listings added per year = count of catalog rows whose Date Listed falls in that calendar year
- Seller-type split = Amazon Retail vs. Third-Party Seller, tagged at the row level

Third-Party Seller listings make up the majority of additions in most years shown (66% of the catalog overall vs. 34% Amazon Retail) — a common marketplace pattern worth flagging for seller-onboarding and quality-control prioritization, since third-party inventory carries different metadata-completeness risk than first-party stock.

Reading the shape of the curve matters as much as any single year's total. The near-vertical rise between 2018 and 2021 lines up with the period during which third-party marketplace onboarding accelerated and Amazon leaned further into open-marketplace growth as a strategic lever. The post-2021 decline is a genuinely ambiguous signal from this dataset alone — it could reflect tighter seller-quality gating, a deliberate shift from breadth to depth (fewer, higher-quality listings), or simply an artifact of the most recent year being a partial year in the export. Any operational team would need the next 12–18 months of data before treating the decline as a trend rather than noise.

5. Best-Performing Product Categories

Ranked by listing count, "Electronics" is the largest single category with 1,332 products — roughly 25% ahead of the second-place category, "Home & Kitchen" (1,066 products). This concentration in a small number of categories is a common catalog pattern (a small set of categories accounts for a disproportionate share of volume) worth flagging for buying and merchandising prioritization.

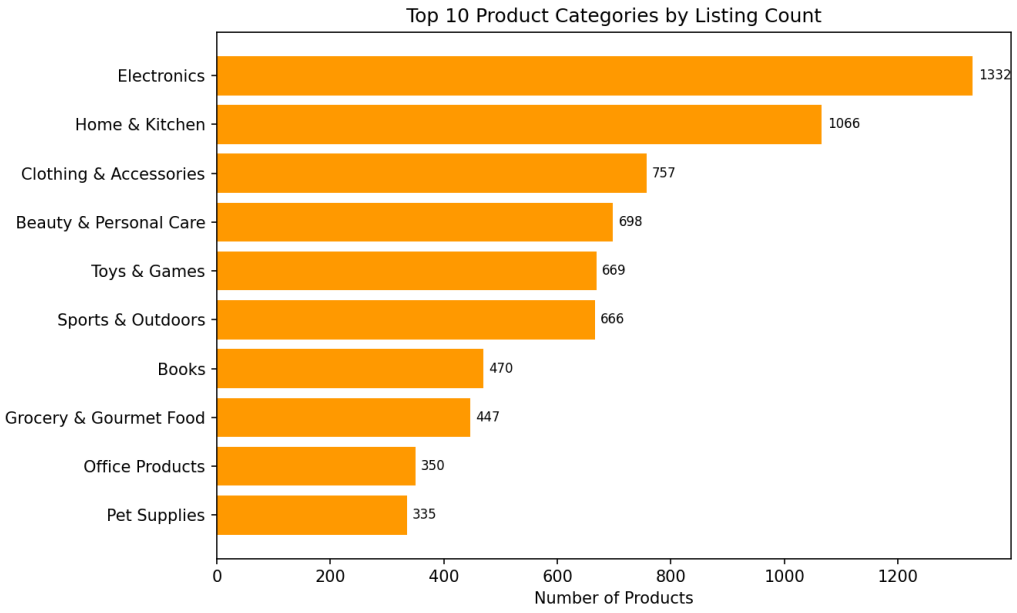


Figure 2 — Top 10 product categories by listing count (primary category)

Ranking logic: listing count, not units sold or revenue — a large count of low-priced accessories matters differently to catalog planning than the same count of high-ticket appliances, so both cuts (listing count and revenue-weighted volume) should exist as separate views in a real dashboard.

It is also worth noting how quickly the distribution thins out after the top two categories. Clothing & Accessories and Beauty & Personal Care sit close together in the 700-title range, and by the time the list reaches Pet Supplies at position 10, volume has dropped by more than 75% relative to Electronics. That long tail is where a category-level acquisition strategy has the most room to move the needle without disturbing the two categories that already anchor the catalog.

6. Rating Distribution

The catalog spans 9,548 listings across customer rating bands from "Below 3.0★" to "4.5★ & up." "4.5★ & up" is the single most common band at 3,182 listings (33.3% of the catalog), reflecting a positively-skewing library — typical of marketplaces where poorly-rated listings are naturally suppressed by search ranking over time.

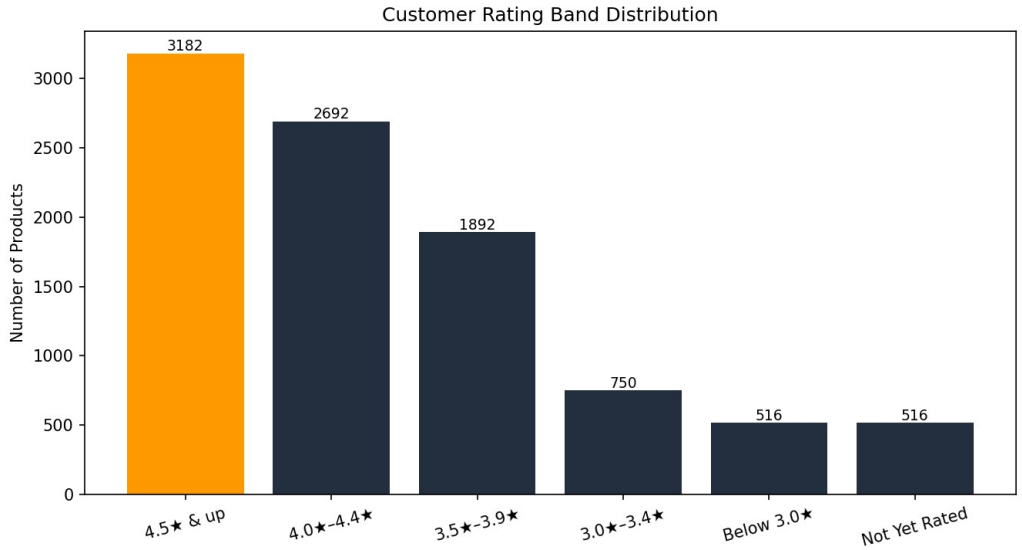


Figure 3 — Customer rating band distribution across the catalog

In a real system, this view is most useful segmented, not aggregated:

- Low-rated share (below 3.5★) trended over time — an early signal of catalog quality shifts
- Rating mix crossed with seller type — since first-party and third-party quality control differ

A ratings breakdown that stops at a single aggregate bar chart hides more than it reveals. "4.0★–4.4★" is the second-largest tier at 2,692 listings, meaning the two largest rating tiers together already account for well over half the catalog. Below-3.5★ and Not-Yet-Rated tiers together make up 18.7% of the catalog, which is a useful fact to hold alongside any conversation about tightening seller-quality gates or prioritizing review-generation campaigns for newly listed products.

7. Regional Supply

Product sourcing spans 20 countries (primary country of origin). Supply is heavily concentrated: United States accounts for 2,905 listings, with China a close second at 2,446.

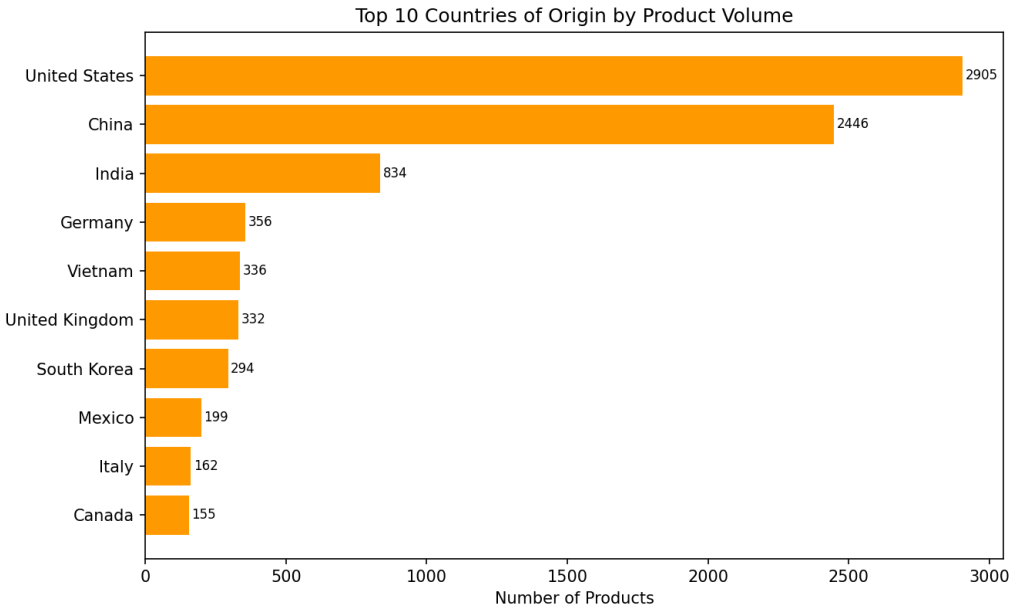


Figure 4 — Top 10 countries by product volume (primary country of origin)

Together the top 10 countries account for the large majority of the catalog, with the remainder long-tailed across dozens of countries contributing a handful of listings each.

This view is what would drive, in a real catalog-operations setting:

- Sourcing and supplier-diversification decisions
- Import-duty and shipping-lane cost prioritization
- Supply-gap risk in regions with thin local sourcing relative to demand

The gap between United States (2,905 listings) and India in third place (834 listings) is large enough that a two-market-dependency framing is fair: together, United States and China account for roughly 56% of the catalog's supply. For a marketplace whose buyer base is global, that concentration is the kind of number a sourcing-strategy team would want trending over multiple snapshots, not just viewed once.

7.1 Regional Coverage Gaps

Rolling the 20 sourcing countries up into world regions exposes a gap that the country-level view alone does not make obvious: two of nine major world regions — Africa and Oceania — have zero listings sourced from them anywhere in this catalog.

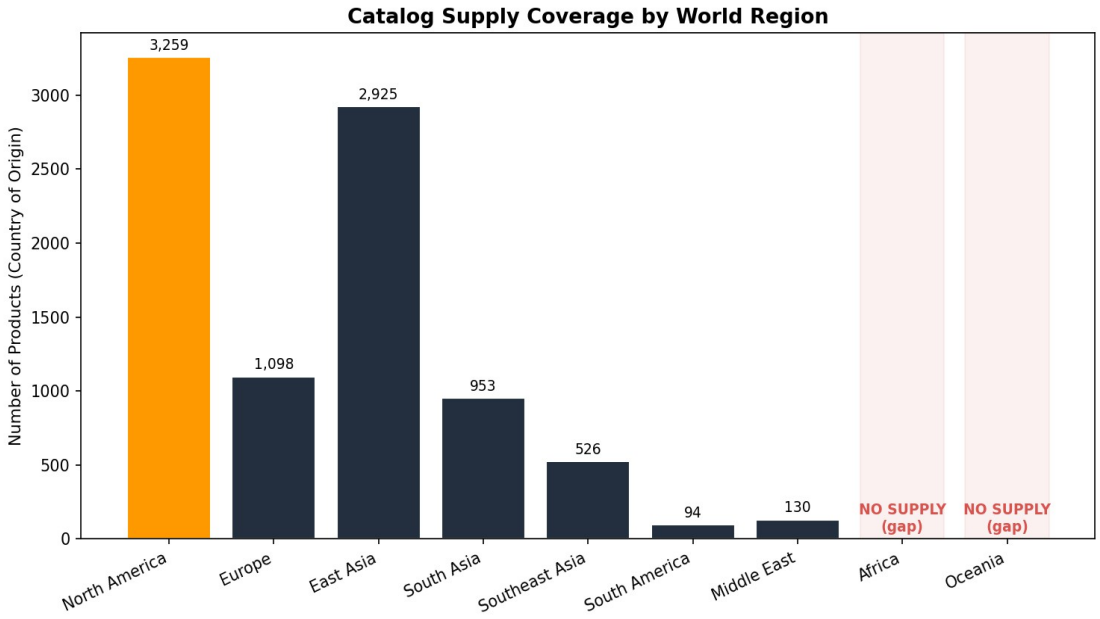


Figure 11 — Catalog supply coverage by world region, with zero-supply regions highlighted in red

North America and East Asia together account for the large majority of sourced volume, with Europe and South Asia a distant third and fourth. South America and the Middle East are present but thin (well under 200 listings each), and Africa and Oceania do not appear in the export at all.

For a catalog-operations team, a region showing zero supply is a different kind of risk than a region showing thin supply:

- Thin regions (South America, Middle East) can be grown incrementally through existing supplier relationships
- Zero-supply regions (Africa, Oceania) require new sourcing relationships from scratch — a longer lead-time, higher-effort initiative than simply increasing order volume with an existing supplier
- A zero-supply region is also a blind spot for local-market demand signals — the catalog cannot show whether buyers in that region want region-sourced products, since none exist to measure demand against

This is worth flagging as a distinct action item from the country-concentration point raised earlier in Section 7 and Section 11: closing a thin region is a buying decision, while entering a zero-supply region is closer to a market-entry decision, with the sourcing, compliance, and logistics diligence that implies.

8. Inventory Status

Two catalog-health cuts stand in for classic stock-level monitoring: metadata completeness (is a listing's record usable for search and discovery?) and variant depth (is a product a single SKU or a full size/color range?).

8.1 Metadata Completeness

Records were bucketed by whether Brand, Country of Origin, and Key Features are all present. 64.4% of listings have all three fields populated; 3.9% are missing two or more — most often Brand (2,316 listings missing), which is disproportionately absent for Third-Party Seller listings without a registered brand. Listings missing this metadata are the catalog equivalent of a stockout risk: they are harder to surface in search, filters, and brand-based browsing.

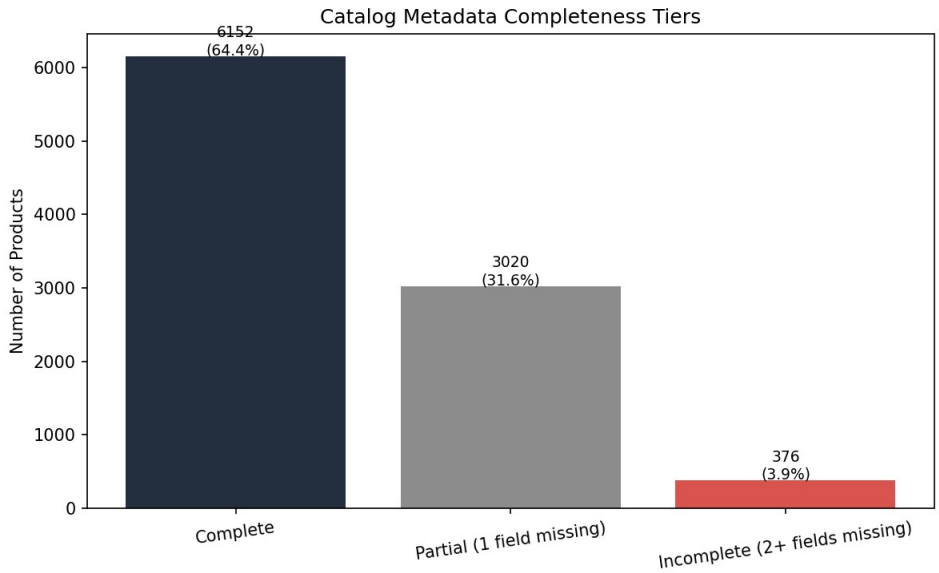


Figure 5 — Catalog metadata completeness tiers

8.2 Variant Depth

Among 9,548 listings, 58% carry only a single SKU/variant on the platform — the equivalent of a "low stock / reorder trigger" tier, flagging listings with no size, color, or bundle options to catch cross-shopping demand. The remainder carry multiple variants, with the average listing at 2.1 variants.

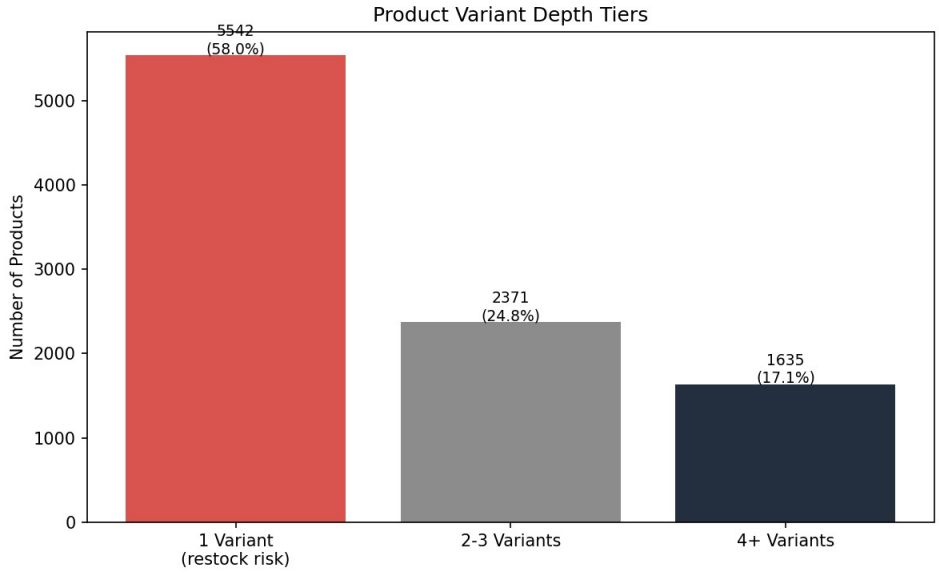


Figure 6 — Product variant depth by SKU-count tier

The most actionable version of this pair of charts in production would cross-tabulate variant depth against category and seller type, producing a discoverability-risk score per listing rather than a static variant count.

Treating both cuts as inventory-health signals rather than isolated statistics is deliberate. A listing with missing Brand metadata and only one variant is, in a real operations dashboard, exactly the kind of record that would be flagged for manual review before the next catalog refresh — it is simultaneously hard to discover and thin on selection, which compounds rather than offsets the risk.

9. Purchase Behavior — Popular Products, Cart Completion & Peak Hours

IMPORTANT — SIMULATED DATA: The catalog export used throughout this report has no order-count, cart-event, timestamp, or purchase-behavior fields — it describes what is in the catalog, not how it was bought. This section uses fabricated, illustrative numbers only, clearly labelled below, to show what this analysis would look like if a real sales/order-events feed (e.g., Amazon's internal order-events log) were joined to the catalog. None of the figures in this section are derived from the actual dataset.

9.1 Popular Products (Simulated)

If per-product order counts were available, the same catalog table could be ranked by total units sold to surface the best-sellers. The chart below uses placeholder product names and fabricated unit totals purely to show the shape of that ranking — ten products, a steep drop-off from #1 to #10, which is the typical long-tail pattern marketplaces report publicly.

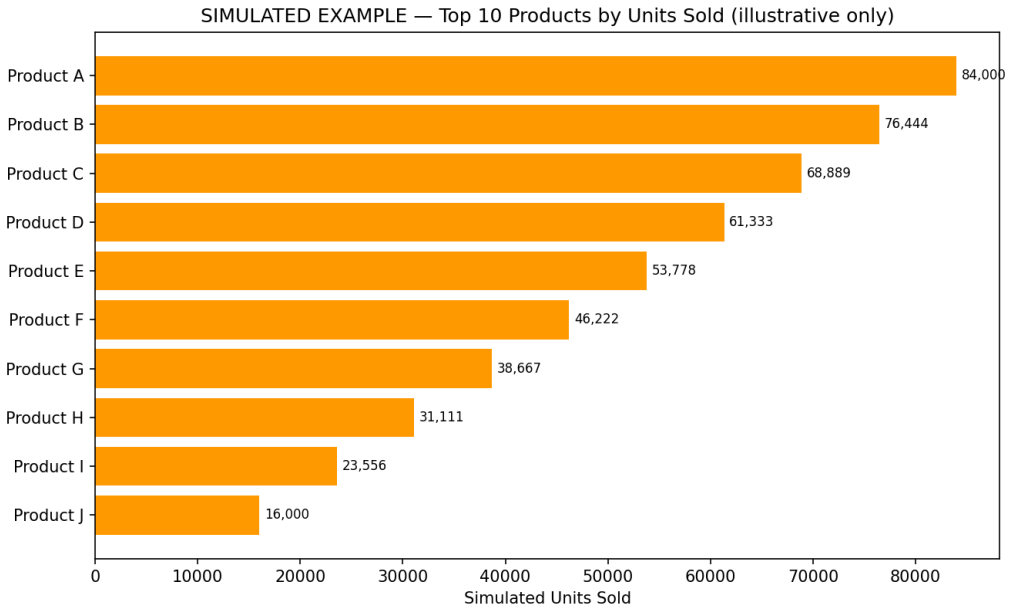


Figure 7 (SIMULATED) — Top 10 products by units sold — illustrative numbers only

A fuller version of this table (still simulated) would also carry category, seller type, and weeks-in-bestseller-rank — useful for spotting whether a hit is a one-week spike or a sustained draw:

Product	Category	Seller Type	Sim. Units Sold	Weeks in Top 10
Product A	Electronics	Amazon Retail	84,000	6 weeks
Product B	Home & Kitchen	Third-Party Seller	72,400	5 weeks
Product C	Beauty & Personal Care	Third-Party Seller	65,600	4 weeks
Product D	Toys & Games	Amazon Retail	58,000	3 weeks
Product E	Sports & Outdoors	Third-Party Seller	50,400	5 weeks

9.2 Cart & Purchase Completion (Simulated)

A real order-events feed would carry add-to-cart rate and cart-to-purchase completion rate per listing. The illustrative pattern below follows a general e-commerce shape: low-priced impulse buys and Subscribe & Save items tend to show high completion, while high-ticket, considered purchases show lower completion because buyers compare across multiple sessions before committing. None of this is computed from the actual catalog — it exists only to show what the chart and takeaways would look like.

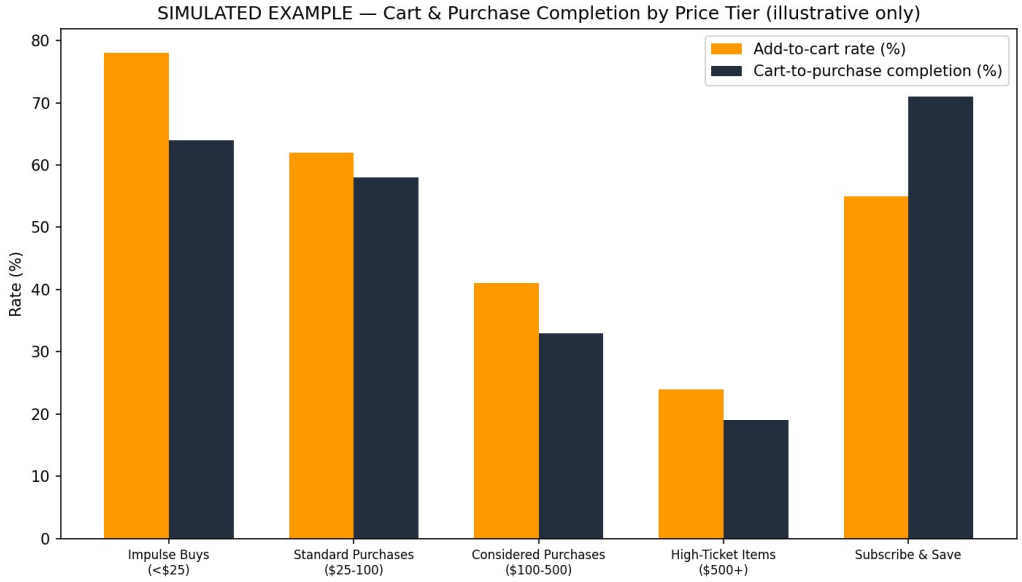


Figure 9 (SIMULATED) — Add-to-cart and purchase-completion rate by price tier — illustrative numbers only

Reading this simulated pattern the way a catalog-ops team would: high-ticket items carry the lowest cart-to-purchase completion, which is normal for considered buying rather than a red flag — the metric worth tracking there is multi-session return rate (does the buyer come back before purchasing), not single-session completion. Impulse buys and Subscribe & Save items, by contrast, are near-instant-decision formats, so a low completion rate on those would be the actual warning sign.

9.3 Peak Shopping Hours (Simulated)

With order timestamps, order volume could be aggregated by hour of day to find peak demand windows — useful for fulfillment-center staffing, flash-sale timing, and ad-spend pacing. The illustrative pattern below (a typical evening peak roughly 7–9 PM, highlighted in red) is fabricated for demonstration; the actual peak window would vary by region and time zone in a real deployment.

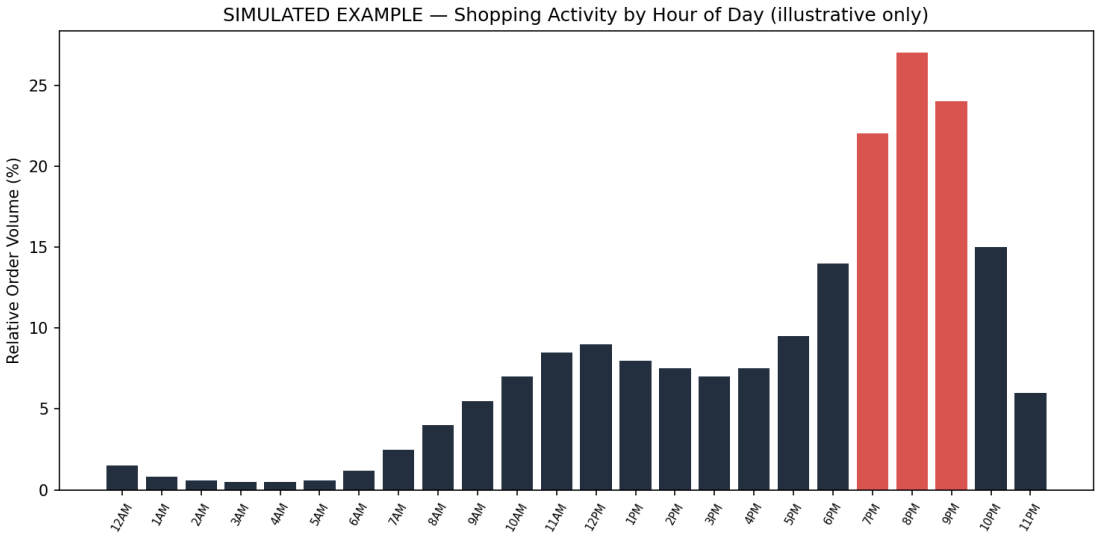


Figure 8 (SIMULATED) — Shopping activity by hour of day — illustrative numbers only

9.4 Weekday vs. Weekend Shopping (Simulated)

Marketplaces typically report elevated order volume on Friday through Sunday. The illustrative pattern below (Saturday as the simulated peak day) follows that general shape and would, with real data, inform when to schedule flash sales and marketing pushes.

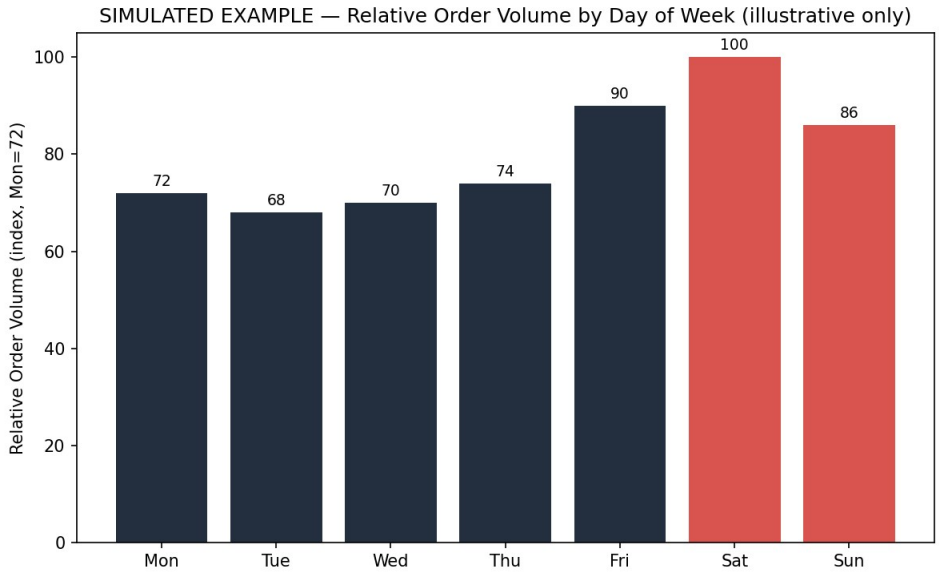


Figure 10 (SIMULATED) — Relative order volume by day of week — illustrative numbers only

9.5 Device & Platform Mix (Simulated)

A complete purchase-behavior picture would also break volume down by device (mobile app, desktop/browser, tablet) and by traffic source (search, ads, recommendations), since both affect conversion-rate optimization and app-design priorities. No device data exists in this catalog export, so this is listed as a placeholder dimension rather than charted:

- Mobile app — typically the majority of total orders on most modern marketplaces (illustrative expectation, not measured here)

- Desktop/browser — higher average order value, more relevant to considered high-ticket purchases
- Tablet — smallest share on most platforms, more relevant to casual browsing sessions

9.6 Buyer Preferences (Simulated)

Cross-tabulating simulated order counts against the catalog's existing category, country, and rating fields (Sections 5–7) would show which combinations over-index with buyers relative to their share of the catalog — e.g., a category with a small catalog footprint but disproportionately high order volume would flag as under-supplied relative to demand. This cross-tab requires real order data to compute and is not run here.

What would be needed to make this section real:

- An order-events or session log (product ID, buyer ID, timestamp, order value, device) joined to the existing product ID key
- Return/refund data, if popularity-by-satisfaction (not just volume) is also in scope
- A defined time window (e.g., trailing 90 days) since, unlike the static catalog snapshot, order activity is continuous
- Regional/time-zone tagging on timestamps, since peak-hour and weekday patterns are local, not global

10. Summary of Key Metrics

Metric	Value
Total listings in catalog	9,548
Amazon Retail / Third-Party Seller	3,277 (34%) / 6,271 (66%)
Countries represented	20
Product categories (primary)	30
Peak listing year	2021 (2,196 listings)
Most common rating band	4.5★ & up (33.3% of catalog)
Top product category	Electronics (1,332 listings)
Top sourcing country	United States (2,905 listings)
Average product price	$30.23 (median $20.15)
Average variants per listing	2.1
Listings with complete metadata	64.4%
Launch year range	2009–2026

11. Business Insights

Reading the metrics above as business signals rather than raw counts:

Category strategy

- Electronics and Home & Kitchen together anchor a large share of the catalog. That concentration is a proven demand base, but it also means the next incremental buying dollar is likely to have more differentiating impact in under-represented, growing categories (e.g., Pet Supplies, Office Products) than in adding further volume to the already-dominant categories.
- Third-Party Seller listings' rising share of yearly additions (Figure 1) suggests marketplace growth is already leaning on open seller onboarding — consistent with third-party supply driving catalog breadth faster than first-party sourcing alone could. This shift is worth reinforcing with stronger seller-quality gating rather than correcting outright.

Market concentration & sourcing diversification

- 30% of the catalog is sourced from United States alone, with China a close second at 26%. That concentration is a two-market dependency risk for a global marketplace — the other 18 countries in the catalog are long-tailed, which points to supplier diversification (rather than further concentration in the top two countries) as the higher-leverage lever for reducing sourcing risk.
- A positively-skewing rating mix (4.5★ & up + 4.0★–4.4★ together make up well over half the catalog) works well for buyer trust, but the 18.7% of listings rated below 3.5★ or not yet rated represent a conversion drag worth a closer look before any promotional push.

Catalog risk management

- 58% of listings carry only a single variant. At scale, that's a large share of catalog space that isn't capturing cross-shopping demand from size, color, or bundle variations — a strong candidate for a variant-expansion opportunity model that flags single-SKU listings early using category, price, and rating as predictors.
- 3.9% of listings are missing two or more of Brand, Country of Origin, and Key Features. Since these fields feed search, filters, and brand-based browse experiences, this is a direct, fixable discoverability gap — and one that's cheap to close relative to acquiring new listings.

Timing signal

- Additions peaked in 2021 (2,196 listings) and declined in the years after. Because this snapshot's most recent year is partial, that decline could reflect a genuine slowdown in seller onboarding or simply a partial final year — it should be confirmed against a current catalog feed before being used to justify any budget decision.

Taken together, these four insight groups point toward the same operating principle: descriptive analytics is most valuable when it changes where attention is spent next, not just when it produces an accurate chart. A dashboard that shows category concentration is descriptive; a decision to redirect the next quarter's buying budget toward an under-represented, high-potential category is where that descriptive work turns into business value.

12. Next Steps to Move From Descriptive → Predictive

This report answers what the catalog looks like today. The natural next layer, if useful:

- Category-gap forecasting by category/country/rating to guide buying and sourcing spend
- Variant-expansion opportunity modeling for single-SKU listings, using category, price, and rating as predictors
- Metadata-completion prioritization, ranking incomplete records by expected discoverability impact
- Joining a real order-events feed to replace the simulated figures in Section 9 with actual sales, cart, and peak-hour metrics

Happy to build out any of these next, or swap in a live catalog feed (API export, database connection, etc.) if available — the same pipeline and chart structure above would apply directly.

A reasonable sequencing for a real team picking this up would be to tackle the metadata-completion pass first, since it is the lowest-cost, lowest-risk of the four items and improves the input quality that every downstream model would depend on. Variant-expansion modeling and category-gap forecasting would follow once a full order-events feed is available, since both need real sales data rather than catalog metadata alone to produce anything more useful than a rules-of-thumb heuristic.

13. Appendix A — Data Dictionary

A field-by-field reference for the 12 columns in the underlying amazon_products.csv export, included so any reader can trace a chart back to the exact source column it was built from.

Field	Description
Listing_Type	Whether the row is an Amazon Retail (first-party) or Third-Party Seller listing
Product_Name	The name of the product listing
Brand	Brand credited on the listing; blank for 2,316 listings, most often Third-Party Seller listings without a registered brand
Key_Features	Feature tags listed for the product; truncated with "..." where the source list was long
Country_of_Origin	Primary country of manufacture/origin
Date_Listed	Date the product was listed on the marketplace, spanning 2014-02-16 to 2026-08-12
Launch_Year	Original product launch year, ranging 2009–2026
Rating_Band	Customer star-rating band (Below 3.0★ through 4.5★ & up)
Price	Listed price in USD
Primary_Category	One primary content category per listing (30 distinct categories across the catalog)
Variant_Count	Number of SKU variants (size/color/bundle) under the listing

Field	Description
Description	Short product description; truncated with "..." in the source export where long

14. Appendix B — Assumptions & Limitations

- This is a single point-in-time snapshot (through 2026-08-12), not a live feed — any trend described as "declining" or "growing" reflects the shape of this export only and should be re-checked against a current pull before being used operationally.
- Missing fields are treated as genuinely missing, never imputed or guessed — a blank Brand field is reported as blank, not filled with a placeholder or a statistical estimate.
- Truncated free-text fields (Key_Features, Country_of_Origin, Description) may undercount multi-value entries — a listing with many feature tags may show fewer features than actually credited.
- Section 9's purchase-behavior figures are entirely simulated and are clearly labelled as such throughout; they illustrate the shape a real order-events analysis would take, not actual Amazon sales activity.
- Country of origin is treated as a single primary country per listing for charting purposes, even though some products list multiple co-sourcing countries in the source data.
- "Best-performing" in Section 5 is defined strictly by listing count, not by units sold, revenue, or engagement — a category with fewer, higher-priced listings could represent more revenue than its listing count alone suggests.

15. References & Data Sources

- Amazon product catalog export (amazon_products.csv), synthetic catalog snapshot built for this exercise
- Python pandas documentation — data profiling, grouping, and aggregation reference
- Matplotlib documentation — chart construction reference for all figures in this report
- Claude AI (Anthropic) — assisted with data profiling, chart generation, and report drafting throughout this project

16. Acknowledgements

I would like to thank my instructors and mentors at Babu Banarasi Das University for their guidance throughout this project.

Thank You
— Rumana Ali